

ONLINE PRODUCT REVIEW SYSTEM USING WEB SCRAPING

Project Report

Submitted By

19BCE7346 : Mamidala Taran

19BCI7012 : Sikhakolli Phanindra

19BCI7068 : Desaboina Deekshitha

Under Guidance of

Dr. Sandipan Maiti

Introduction :

- After pandemic we have accelerated the shift towards a more digital world. These days online shopping is very easy and comfortable for buying products online.
- Like around 2 billion people in the world use e-commerce sites to buy daily necessities and things online. Almost 67 percent of shopping is done on online ecommerce sites.
- Day by day the demand for online shopping is rapidly increasing and the market is growing by 25% to 30% every year.
- Amazon, flipkart is one such widely known E-commerce website and it is being used by people worldwide.



Motivation :

- Getting public opinion from the text reviews from various websites and analysing becoming a trending study these days.
- People would love to know the public opinion while purchasing the products online.
- These insights given by customers or public can be used to improve the product and understand what the producer missed and what can be changed
- This also helps other customers to search products easily and assist others in making a decision about that product
- The approach used is to rate a product based on its technical specification by analysing large number of user reviews which are extracted dynamically from several top e-commerce websites

Objectives :

- The Main objective is to benefit the customers and assist them in choosing the right product while shopping.
- This project would carry out sentiment analysis of online product shopping based on reviews and predict whether this review is 'Positive', 'Neutral', 'Negative'.
- Here the Sentiment analysis is a text analysis method that detects polarity within text, whether a whole document, paragraph, sentence, or clause for getting the required output.
- This would also help the company in knowing their product from customer view and would improve product for its best.

Hypothesis :

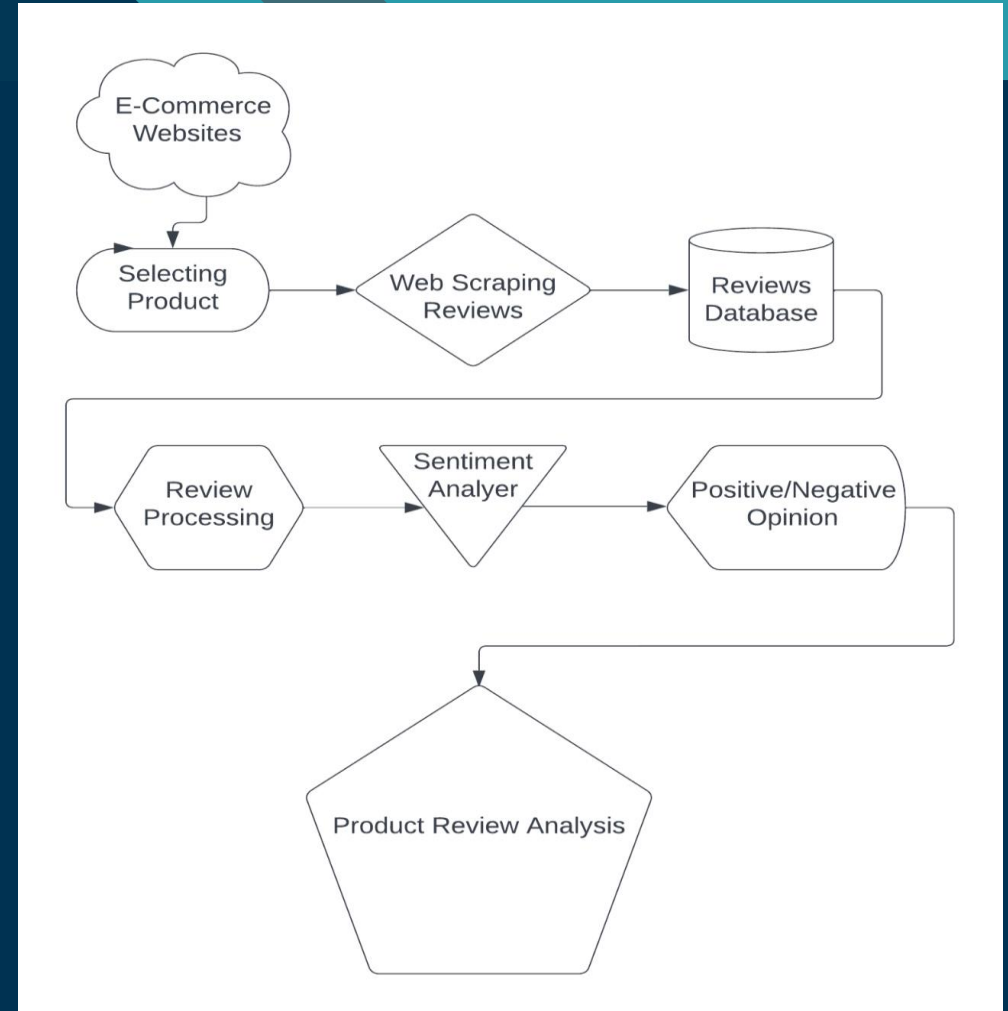
- The reason for selecting this project is because understanding people's emotions is very important for businesses since customers could express their thoughts and feelings for that respective product.
- By automatically analyzing customer feedback using scraping, from product review from their customer experience, brands can listen attentively to their customers, and tailor products and services to meet their needs.
- Like using sentiment analysis to automatically analyze the required limit reviews about your product
- This could help you discover if customers are happy about your pricing plans and customer service.

Problem Survey :

- These days, many users who are using websites, blogs, online shopping tends to review the products they used. These reviews were taken into consideration by other users during their search for products.
- we purchase products online, making all the information available at our fingertips. As the access to information becomes easier, more and more consumers will seek product information from other consumers apart from the information provided by the seller.
- So, In this project the dataset is collected from the online shopping sites - Amazon or Flipkart in large amount.
- Initially, these reviews must be pre-processed in order to remove the unwanted data such as stop words, be verbs, punctuations, and conjunctions. Once, the pre-processing is over the trained dataset is classified from Machine Learning techniques.

Subject Knowledge :

- Scrapping customer product reviews on various websites featuring various products from Amazon or Flipkart.
- Will be using some preprocessing techniques on dataset and perform ml techniques.
- We will Analyse sentiment on dataset from document level (review level).
- And will be using Machine learning and Deep Learning for performing sentiment Analysis.
- The categorization or classification is done based on opinion sentiment into-Positive and Negative



Time Plan :



- In this project we have presented our work model on sentiment analysis of product reviews on E-Commerce site Amazon, Flipkart. We will be doing feature based classification of reviews.
- This would benefit the customers and assist them in choosing the right product for their needs.
- As future work we propose to offer a summary of reviews for the products and recommend the products based on the features that the user is interested in.

References :

- <https://www.analyticsvidhya.com/blog/2021/06/web-scraping-with-python-beautifulsoup-library/>
- <https://ieeexplore.ieee.org/abstract/document/9058128>
- <https://ieeexplore.ieee.org/abstract/document/8376299>
- https://www.researchgate.net/publication/339072438_A_Review_of_Online_Product_Reviews